

Deterring a Small Collusion Is All You Need

Yotam Gafni*

Weizmann Institute of Science
Rehovot, Israel

Abstract

Transaction Fee Mechanisms (TFMs) study auction design in the Blockchain context, and emphasize robustness against miner and user collusion, moreso than traditional auction theory. [7] introduce the notion of a mechanism being *c*-Side-Contract-Proof (*c*-SCP), i.e., robust to a collusion of the miner and *c* users. Later work [6, 15] shows a gap between the 1-SCP and 2-SCP classes. We show that the class of 2-SCP mechanisms equals that of any *c*-SCP with $c \geq 2$, under a relatively minor assumption of consistent tie-breaking. In essence, this implies that any mechanism vulnerable to collusion, is also vulnerable to a small collusion.

CCS Concepts

• **Theory of computation** → **Computational pricing and auctions**; **Algorithmic mechanism design**; **Solution concepts in game theory**.

Keywords

Collusion in Auctions, Blockchains, Transaction Fee Mechanisms

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1 Introduction

Transaction Fee Mechanisms (TFMs) is the name given to the class of mechanisms responsible for the ordinary operation of a blockchain system, choosing which transactions are confirmed at each issued block. In Bitcoin, this is done by having each transaction commit to a fee that it pays if confirmed. This is in essence a pay-as-bid auction, and a line of work since [20, 23] has considered alternative auction designs. Importantly, this has not remained a theoretical exercise, and major blockchain protocols have adopted different mechanisms, most notably Ethereum, which adopted the EIP-1559 mechanism (in essence, a first-price with a reserve that is fully burned) [27].

TFMs have several unique characteristics that distinguish them from traditional auctions [33]: (i) They allow for payments to be “burned”, i.e., the user pays but the miner (which takes the role of the auctioneer) does not receive the full payment, (ii) Their research is more focused on *robustness* to different forms of attacks, rather than

just incentive-compatibility w.r.t. the bidders, and welfare/revenue maximization within this framework, which is the standard classic approach to auctions [24, 34]. In particular, the issue of *collusion-resistance* has taken an important role in TFM research [7, 28]. Even in traditional auctions, practitioners have long emphasized the importance of collusion-resistance. The following is an emblematic quote:

“The most important features of an auction are its robustness against collusion and its attractiveness to potential bidders. Failure to attend to these issues can lead to disaster.” [22]

However, this takes maybe even greater precedence in Blockchain systems, because of their online and ephemeral nature. In an anonymous, unregulated environment, bidders may worry less about facing prosecution for cartel behavior, as they have faced many times in traditional auctions [26]. Thus, TFM research strives to make collusion-resistance a built-in part of its design.

We now arrive at the topic at the center of this paper. [11] raise the question, supported by influential work in economic theory [32], of whether collusion is a real risk in TFMs. The argument goes, that since in some systems block times are quick (e.g., 12 seconds in Ethereum), it is hard to not only establish a collusion, but also coordinate, through possibly several rounds of communication, how each agent should partake in it. Both in traditional auctions and in TFMs, there is some support in theory and empirics that the *smaller* the collusion, the *easier* it is to establish it, coordinate it, and avoid its detection. Thus, it is important to know whether mechanisms are robust to *small-size* collusion.

Our work does exactly that: We consider the popular collusion notion of *c*-SCP, introduced by [7]. The notion considers whether a coalition of the miner and up to *c* bidders is able to increase its utility by changing bids, as well as using manipulations by the miner (dropping bids, or adding shill bids). The *c* parameter naturally establishes a hierarchy: Clearly, the class of 1-SCP mechanisms (i.e., these robust to a collusion of the miner and at most a single user) is the widest class, while the class of SCP (used as shorthand for *n*-SCP, i.e., a collusion of the miner and any number of bidders) is the narrowest. Moreover, an example of [6] shows a gap between the class of 1-SCP and 2-SCP mechanisms. It is thus natural to ask,

Question: Is there a separation between each *c*-SCP and $(c + 1)$ -SCP class?

We answer this question in the negative, and show that other than the 1-SCP and 2-SCP gap, there are no more gaps, and in terms of mechanism classes, 2-SCP = SCP. Moreover, if we can find an arbitrary collusion against a mechanism, then it is reducible in poly-time to a simple one, using our explicit reduction. Thus, at least in terms of the need to involve many parties in the collusion, we show that this is not an issue. We detail remaining hurdles for the colluders in the discussion.

*Equal contribution.



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1.1 Related Work

1.1.1 The Core TFM Framework and Extensions. Considering blockchain transaction fee mechanisms as an auction design problem was studied by [20, 23]. The early emphasis was on *miner* incentive-compatibility (MIC), i.e., robustness to manipulations by the miner such as dropping bids, and introducing fake bids. [28] was the first to introduce collusion-resilience as the centerpiece of TFM design, through the notion of OCA-proof. OCA-proof is a notion that requires the mechanism to be welfare-maximizing for the *grand coalition*. [7] introduce the *c*-SCP notion, which captures a more game-theoretic group deviation notion, which we study in this paper. The focus of the axiomatic literature that followed was on showing whether a “dream” TFM, that combines collusion-resilience, as well as the user and miner incentive-compatibility notions, exists. The overall conclusion is that the answer is no (To mention some results, no user incentive-compatible (UIC) + 1-SCP mechanism [7] exists, no UIC+MIC+OCA-proof deterministic mechanism, scalable randomized one, or randomized where the collusion knows the random coins exists [6, 12]), although some open questions remain. In particular, a non-scalable randomized mechanism where the collusion is oblivious to the random coins may still exist, though its welfare is bounded away from optimal [12]. However, one might be interested in a subset of the restrictions. For example, if we think that regular users are ephemeral and have bounded rationality, we may omit the user incentive-compatibility requirement. One reason to think so is the common use of fee estimators and calculators by users [36]). With only a subset of restrictions, there are characterizations of the classes of mechanisms that implement them [12]. Importantly, we are not aware of a past work that examined the notion of SCP in isolation, and drew conclusions and characterizations from it alone, and this is a novelty of our work. In particular, our Corollary 4.3 characterizes single-item 1-SCP mechanisms as exactly the class of EIP-1559 mechanisms, which validates its inception, motivated by robustness to collusion [5]. [12] has a characterization of UIC and Global-SCP single-item mechanisms as generalized EIP-1559 mechanisms. However, this invokes both a collusion notion and single-agent incentive-compatibility, and shows a class that strictly contains EIP-1559.

Following the works that showed impossibility, several works [9, 16, 17] consider refined collusion notions, that emphasize incentive tensions among the colluders. To counter that, [21] introduce the idea of *smart collusion*, where colluders can impose penalties on defectors.

Beyond the core framework, the study of TFMs was extended in many directions: (1) Considering time preferences [1, 11, 14, 25], (2) Considering the problem of Maximal Extractable Value (MEV), i.e., incentive problems that may arise in certain blockchain use-cases where the order of the transactions in the block matters [2], (3) Considering more advanced blockchain architectures, such as DAGs and PBS [3, 18]. We emphasize that our work is in the core TFM setting, and does not consider these variations.

In the full version, we overview additional related work that details the theory and empirical evidence of collusion, both in Blockchains and traditional auctions.

1.2 An Overview of Our Proof

Our proof is constructive and presents a reduction from a given general collusion to a 2-collusion. First, we make sure the collusion is in what we call “Canonical Ordering”. This means that bidders in the collusion do not change their relative order, in terms of value, before and after the collusion. This is possible since a collusion can always have even better utility by allowing higher-value agents to be confirmed. Keeping the canonical ordering throughout our reduction is useful in making the problem one-dimensional (‘how many colluding bidders are confirmed?’) instead of multi-dimensional (‘is each of the colluding bidders confirmed?’). The main idea is to decompose the collusion into steps, where at each step only one bidder changes their bid. The goal is to isolate one step which is in itself a beneficial collusion. However, this has to be done carefully. Two important insights allow the decomposition:

(i) To maintain 2-SCP, there must be a monotonicity of the number of confirmed bidders in whether a bidder has increased / decreased their bid. If a bidder in the collusion increases their bid, the number of confirmed bidders must weakly grow, and vice versa. In some sense, this is like incentive-compatibility for a single bidder, but applied to the set of all the colluders. There are however some subtleties, since the miner is also part of the joint utility of the collusion, and also we need to only use the 2-SCP condition (and not the more general SCP).

(ii) We aim to have some kind of a telescopic argument: If you have a decomposition of the collusion into one-by-one steps, then one of the steps is a collusion in itself. For this to work, the changes at each step should accumulate in a way that corresponds to the shift from the original A to B . However, the changes may be incorrectly accounted for in this shift: Every time a bidder changes their bid and is confirmed, the difference in values is accounted into the sum, but these may be different than the bidders confirmed in B , whose difference of values is accounted for in the collusion. This goes both ways: There may be bidders confirmed in the decomposition that are not confirmed in B , and vice versa. Thus, we need to make sure, as Lemma 3.10 requires, that “increasing and ultimately unconfirmed” bidders are not confirmed throughout the decomposition, and “decreasing and ultimately confirmed” are confirmed in the decomposition. It turns out that this is guaranteed with a certain decomposition, where the “increasing and confirmed” bidders move first, then all decreasing bidders move, and finally the “increasing but unconfirmed” bidders move. We nick-name this construction “The Salsa Dance Decomposition” because of the resemblance to the forward-backward-forward nature of the basic Salsa dance step.

We are thus able to reduce any collusion to a collusion where a single bidder moves, and all other parties to the collusion are passive beneficiaries of this change. It then remains to isolate a single beneficiary of the change, and have the mover and the beneficiary would constitute the collusion, together with the miner. This in itself is not trivial since the 2-collusion needs improve its joint utility: For example, you could have a “mover” that loses utility of 10 due to the move, and 3 beneficiaries that gain 4 each. While it is easy to isolate a single beneficiary (take any of them), this will not suffice for a 2-collusion. This again requires making small steps, this time w.r.t. how the mover changes their bid. We show that by gradually increasing the bid we can isolate a change where the

mover has a small utility loss, but at least one of the beneficiaries has a larger utility gain.

2 Model and Preliminaries

We follow the standard model used by the literature for Transaction Fee Mechanisms (TFMs) [7, 23, 29, 31]. The model assumes a multi-unit, unit-demand setting, i.e., bidders have a fixed-size transaction, and do not care about specific placement in the block. We interchangeably use the terms *auction*, *mechanism*, or transaction fee mechanism (TFM) to refer to the object of our discussion, depending on what is most appropriate in the context. Missing proofs appear in the full version of the paper.

2.1 Transaction Fee Mechanisms

A TFM consists of a *confirmation rule* a , a *payment rule* p and a *burning rule* β .

Confirmation rule. A confirmation rule defines the mechanism's confirmation of transactions to the upcoming block, as intended by the mechanism's designers.

DEFINITION 2.1 (CONFIRMATION RULE). A deterministic *confirmation rule* $a : R_+^n \rightarrow \{0, 1\}^n$ defines which transaction should be included in the upcoming block. Generally, a confirmation rule may be randomized, i.e., $a : R_+^n \rightarrow \{\Delta(0, 1)\}^n$.

Payment & burn. The payment and burn rules respectively define the amount of fees paid by transactions when included in a block, and how much of each payment is “burnt” and taken out of circulation instead of given to the miner.

DEFINITION 2.2 (PAYMENT RULE). The payment rule $p : R_+^n \rightarrow R_+^n$ receives bids $b_1, \dots, b_n$, and outputs a payment vector with a payment for each bidder i .

DEFINITION 2.3 (BURNING RULE). The burning rule $\beta : R_+^n \rightarrow R_+^n$ in auction A receives bids $b_1, \dots, b_n$, and outputs a burn vector with a burn for each bidder i , meaning the amount of funds out of b_i 's payment that is taken out of circulation.

DEFINITION 2.4 (AGENT UTILITIES). Let auction $A = (a, p, \beta)$. Consider n bidders with true valuations $\mathbf{v} \in R_+^n$ and n' bids $\mathbf{b} \in R_+^{n'}$. W.l.o.g., $\mathbf{b}$ is indexed the same as $\mathbf{v}$, meaning that the first n bids in $\mathbf{b}$ correspond to the bids in $\mathbf{v}$ (possibly as 0 if omitted), thus w.l.o.g. let $n' \geq n$. We consider the $n' - n$ bidders that are in $\mathbf{b}$ but not in $\mathbf{v}$ as the miner's “fake bids”, and denote them by $\mathbf{b}_F$. Then, the utilities of the various agents are defined as follows.

- Bidder Utility:

$$u_i(b_i, \mathbf{b}_{-i}; v_i) \stackrel{\text{def}}{=} v_i \cdot a(b_i, \mathbf{b}_{-i}) - p(b_i, \mathbf{b}_{-i}). \quad (1)$$

- Miner Utility:

$$\begin{aligned} u_{\text{miner}}(\mathbf{b}; \mathbf{v}) &\stackrel{\text{def}}{=} \sum_{i=1}^n (p(b_i, \mathbf{b}_{-i}) - \beta(b_i, \mathbf{b}_{-i})) + \sum_{i=n+1}^{n'} u_i(b_i, \mathbf{b}_{-i}; 0) \\ &= \sum_{i=1}^n (p(b_i, \mathbf{b}_{-i}) - \beta(b_i, \mathbf{b}_{-i})) - \sum_{i=n+1}^{n'} \beta(b_i, \mathbf{b}_{-i}). \end{aligned} \quad (2)$$

It will sometimes be convenient to use the shorthand

$$u_C(X; Y) \stackrel{\text{def}}{=} \sum_{i \in C} u_i(X; Y).$$

We assume an auction is individually rational: For any $\mathbf{b} \in R_+^n$ and for each bidder i ,

$$u_i(b_i, \mathbf{b}_{-i}; b_i) \geq 0. \quad (3)$$

We also assume it is burn-balanced, i.e., the burn of each bidder does not exceed their payment:

$$p(b_i, \mathbf{b}_{-i}) \geq \beta(b_i, \mathbf{b}_{-i}) \geq 0. \quad (4)$$

We assume that an auction is anonymous, so that a permutation over the bids results in a permutation over the confirmed bids. For any bid vector $\mathbf{b} \in R_+^n$, permutation $\pi \in S_n$, and rule $x \in \{a, p, \beta\}$, then $x(\pi(\mathbf{b})) = \pi(x(\mathbf{b}))$.

We say a bidder is a zero-utility bidder if $u_i(b_i, \mathbf{b}_{-i}; b_i) = 0$: I.e., it is a bidder that is either unconfirmed, or pays their bid.

We make the following tie-breaking assumption:

DEFINITION 2.5. *Consistent Tie-Breaking towards zero-utility bidders, w.r.t. unconfirmed bidders.*

Consider a bidder j and bids $\mathbf{b}_{-j}$ so bidder j is unconfirmed both when bidding b_j , and when bidding b'_j . Call the former setting A and the latter B .

Then any bidder i that is zero-utility in both settings A and B , has the same confirmation status in both.

This definition is natural in the sense that it holds for reasonable auction formats.

Example 2.6. Consider a single-item posted-price auction with price p and lexicographic tie-breaking, i.e., for bidders $b_1, \dots, b_n$, the lowest index bidder with $b_i \geq p$ wins the item and pays p . Then this tie-breaking is consistent towards zero-utility bidders, w.r.t. unconfirmed bidders. Let us unpack the definition directly: Consider some unconfirmed bidder j . It either has a higher index than the confirmed bidder, or $b_j < p$. Consider it changes its bid and remains unconfirmed (then, in the latter case, we have $b'_j < p$). Now consider a zero-utility bidder i : If i is unconfirmed, then i remains unconfirmed. If i is confirmed, then it must be that $b_i = p$, and it remains confirmed in both settings.

Alternatively, consider a second-price auction with lexicographic tie-breaking. I.e., the highest bidder wins, and if there are several equal highest bids, the lowest indexed one wins. Similarly, this is consistent towards zero-utility bidders, w.r.t. unconfirmed bidders.

We note that the property holds for specially-constructed formats, such as the discount auction of [6].

DEFINITION 2.7 (c-SIDE-CONTRACT-PROOF (SCP), WITH ACTIVE / PASSIVE MINER). For a bid vector $b_1, \dots, b_n$, a side-contract between a miner and a coalition of bidders C is such that bidders in the coalition $i \in C$ may change their bids to b'_i . In the active model, the miner may also omit bids (whether by coalition members or not), and add $n' - n$ fake bids. We denote the original honest setting A and the setting adjusted by the collusion B . We use the term **setting** to mean a certain combination of bids. We say that a collusion is beneficial if the joint utility of the miner and the bidders in C is greater in B than in A , given that their true valuations are from A .

$$u_{\text{miner}}(A; A) + \sum_{i \in C} u_i(A; A) < u_{\text{miner}}(B; A) + \sum_{i \in C} u_i(B; A). \quad (5)$$

A TFM is *c-side-contract-proof (SCP)-proof* if a coalition of a miner and a coalition C of the bidders with $|C| \leq c$ cannot increase their aggregate utility by deviating from the honest protocol.

Notice that in the passive miner model, since the miner does not omit or introduce fake bids, we may denote $u_{\text{miner}}(X)$ instead of $u_{\text{miner}}(X; Y)$, as Y only matters in the miner's utility to distinguish the fake bids. We sometimes refer to a possible arrangement between a coalition and a miner as a collusion, distinguishing it from a *beneficial* collusion, which satisfies the above condition that their joint utility improves. We also say there exists a *c-SC* (*c-Side-Contract*) when there is a beneficial collusion with c bidders.

3 The class of 2-SCP mechanisms equals the class of SCP mechanisms

As observed by [15], the Discount auction of [6] shows a nice separation between 1-SCP and 2-SCP. We show that there are no other gaps between *c-SCP* and $(c + 1)$ -SCP mechanisms. We start by noting that we can restrict ourselves to the passive miner model of collusion.

Lemma 3.1. *If a beneficial collusion exists against a mechanism in the active miner model, then it is either not 1-SCP, or there exists a beneficial collusion against it in the passive miner model.*

Lemma 3.2 (2-SCP $\implies$ a subset of the highest bids is confirmed). *If a, p, β is 2-SCP, then for a bids vector $b_1, \dots, b_n$ sorted in descending order, it must be that $\exists 0 \leq \ell \leq n$ so that all bids up to and including ℓ are confirmed, and all other bids are not.*

PROOF. Consider if for some two bids $b_i > b_j$, the latter is confirmed while the former is not, and consider the 2-SC where bidder i bids b_j and bidder j bids b_i . The miner utility is unchanged, the sum of utilities for agents i, j increases by $b_i - b_j$, so it is a 2-SC. $\square$

Lemma 3.3 (Canonical Ordering). *Consider a beneficial collusion of a miner and a coalition C so that each $i \in C$ changes their bid from b_i to b'_i (possibly, with $b_i = b'_i$). Then, we can rearrange the collusion so that the highest bid among $\{b_i\}_{i \in C}$ goes to the highest bid among $\{b'_i\}_{i \in C}$, and so on, and this is a beneficial collusion.*

PROOF. The new collusion we define has the same payments and burns, since the mechanism is oblivious to the underlying values before the collusion. Moreover, by Lemma 3.2, the top ℓ bids among b'_i (for some ℓ) are confirmed. The value after the collusion is maximized if the underlying values before the collusion in these confirmed spots are the top ℓ values in b_i , which is the case after the rearrangement. $\square$

Lemma 3.4 (2-SCP $\implies$ A Form of Non-Bossiness). *Consider a bidder i bidding b_i in A and b'_i in B , and this is the only change in bids between the settings. Assume bidder i is unconfirmed in both settings. Then, the number of non zero-utility bidders in B is the same as in A .*

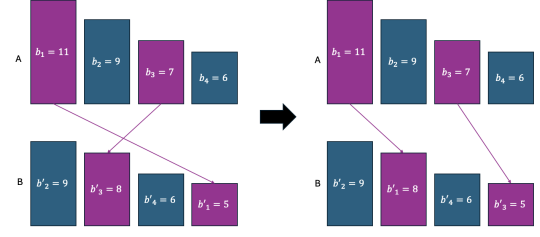


Figure 1: A demonstration of Lemma 3.3. Given a collusion where bidders change their order after the collusion, we can find a collusion where the order is maintained.

PROOF. Assume towards contradiction there is a bidder j that is a zero-utility bidder in A and a non-zero-utility bidder in B .

First, consider a collusion between the miner and bidder i . Then, by the 1-SCP condition from A to B and vice versa,

$$\begin{aligned} u_{\text{miner}}(A) &= u_{\text{miner}}(A) + u_i(A; A) \\ &\geq u_{\text{miner}}(B) + u_i(B; A) = u_{\text{miner}}(B), \\ u_{\text{miner}}(B) &= u_{\text{miner}}(B) + u_i(B; B) \\ &\geq u_{\text{miner}}(A) + u_i(A; B) = u_{\text{miner}}(A), \end{aligned}$$

which yields $u_{\text{miner}}(A) = u_{\text{miner}}(B)$.

Next, consider a collusion between the miner and bidders i, j . Then, by the 2-SCP conditions from A to B and vice versa,

$$\begin{aligned} u_{\text{miner}}(A) &= u_{\text{miner}}(A) + u_i(A; A) + u_j(A; A) \\ &\geq u_{\text{miner}}(B) + u_i(B; A) + u_j(B; A), \\ u_{\text{miner}}(B) &= u_{\text{miner}}(B) + u_i(B; B) + u_j(B; B) \\ &\geq u_{\text{miner}}(A) + u_i(A; B) + u_j(A; B), \end{aligned}$$

which yields:

$$u_{\text{miner}}(A) \geq u_{\text{miner}}(A) + u_j(A; B) + u_j(B; A), \quad (6)$$

i.e.,

$$u_j(A; B) + u_j(B; A) \leq 0.$$

However, notice that for bidder j , since it does not change its bid between A and B , and it is a zero-utility bidder in A and non-zero-utility bidder in B , it holds that:

$$u_j(A; B) + u_j(B; A) = 0 + b_j - p_i^B > 0,$$

in contradiction. $\square$

Lemma 3.5 (Increase Monotonicity). *If a collusion has a single unconfirmed bidder ℓ that increases their bid from A to B , so that it remains lower than the lowest confirmed bidder in A , then the set of confirmed bidders weakly increases from A to B .*

PROOF. Consider a collusion of the miner and all agents in a coalition C , and some single bidder $\ell \in C$ so that $b'_\ell > b_\ell$. Let A be the setting before the collusion (i.e. with all agents bidding b) and B the setting after the collusion.

If bidder ℓ is a non-zero-utility bidder at B , then ℓ is confirmed in B , and by Lemma 3.2, all bidders confirmed in A are confirmed in B as well.

Assume towards contradiction that bidder ℓ is a non-zero-utility bidder at A but is a zero-utility bidder at B . By the 1-SCP condition for the miner and agent ℓ , w.r.t. going from A to B and vice versa, it holds that

$$u_{\text{miner}}(A) + b_\ell - p_\ell^A \geq u_{\text{miner}}(B) \geq u_{\text{miner}}(A) + b'_\ell - p_\ell^A,$$

which is a contradiction as $b'_\ell > b_\ell$.

If bidder ℓ is a zero-utility bidder at both A and B :

- If bidder ℓ is confirmed in B , then condition (i) is satisfied.
- If bidder ℓ is unconfirmed in both A and B , then by Lemma 3.4 the set of non-zero-utility bidders remains the same at A and B . By consistent tie-breaking, so do zero-utility bidders, and so condition (ii) is satisfied.
- If bidder ℓ is confirmed in A and unconfirmed in B , consider the 1-SCP condition from A to B :

$$u_{\text{miner}}(A) = u_{\text{miner}}(A) + u_i(A; A) \geq u_{\text{miner}}(B) + u_i(B; A), \quad (7)$$

Then, combine it with the 1-SCP condition from B to A :

$$u_{\text{miner}}(B) = u_{\text{miner}}(B) + u_i(B; B) \geq u_{\text{miner}}(A) + u_i(A; B)$$

$$\begin{aligned} \text{Eq. 7} \quad &\geq u_{\text{miner}}(B) + u_i(B; A) + u_i(A; B) = u_{\text{miner}}(B) + b'_i - p_i^A \\ &> u_{\text{miner}}(B) + b_i - p_i^A = u_{\text{miner}}(B), \end{aligned}$$

which is a contradiction. $\square$

A similar (but opposing) statement holds for decreasing bids.

Lemma 3.6 (Decrease Monotonicity). *If a collusion has a single confirmed bidder ℓ that decreases their bid from A to B , so that it remains higher than the highest unconfirmed bidder in A , then the set of unconfirmed bidders weakly increases from A to B .*

DEFINITION 3.7 (THE “SALSA DANCE” DECOMPOSITION.). *Let A and B be two bid vectors. Denote U the bidders increasing their bid from A to B , and D the bidders decreasing their bids. Let W_B be the confirmed bidders in B . Let $U_I = U \cap W_B$, $U_O = U \cap L_B$. Let $D_I = D \cap W_B$, $D_O = D \cap L_B$.*

We construct SEQ in the following way:

1. $\text{SEQ} = \{A\}$.
2. For each bidder $i \in U_I$ in descending order of their bids in A , let X be the current last vector added to SEQ , then add $\text{SEQ} = \text{SEQ} \cup \{b'_i, X_{-i}\}$.
3. For each bidder $i \in D_I$ in increasing order of their bids in A , let X be the current last vector added to SEQ , then add $\text{SEQ} = \text{SEQ} \cup \{b'_i, X_{-i}\}$.
4. For each bidder $i \in U_O$ in descending order of their bids in A , let X be the current last vector added to SEQ , then add $\text{SEQ} = \text{SEQ} \cup \{b'_i, X_{-i}\}$.

The following lemma shows that the canonical ordering is maintained throughout the decomposition.

Lemma 3.8 (Order-Maintaining Decompositions). *Consider a canonical ordering collusion of a miner and a coalition C so that each $i \in C$*

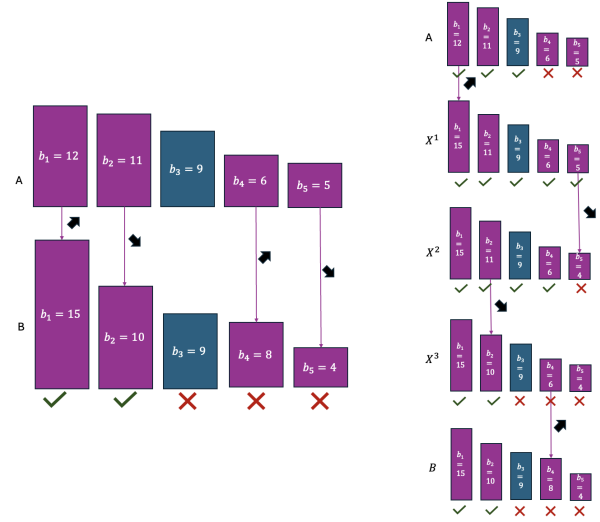


Figure 2: The “Salsa Dance” Lemma. We separate the bidders into U_I, D, U_O , and let each bidder change their bid one-by-one following Definition 3.7. Lemma 3.10 restricts the possible confirmations of each step in the decomposition, and the figure shows one such valid sequence of confirmations.

changes their bid from b_i to b'_i . Let U be the bidders that raise their bid, and D the bidders that decrease their bid. Consider any decomposition where the subsequence of changes of bidders in U is in descending order (higher bidders are earlier in the sequence), and the subsequence of changes of bidders in D is in ascending order. Then, every two subsequent bid vectors, as a (not necessarily beneficial) collusion of the miner and the coalition C , are in canonical ordering.

Before we prove the main structural lemma for the “Salsa Dance” decomposition, we demonstrate that the specific choice of decomposition matters for its success.

Example 3.9. Consider an auction with $k = 2$ items with the allocation rule that the two highest bidders v_1, v_2 are both confirmed if and only if $v_1 + v_2 \geq 2$. Regarding payments, v_2 pays their bid, and v_1 pays the bidders’ average $\frac{v_1 + v_2}{2}$. All payments are burned. Consider bidders with true values $b_1 = 1, b_2 = 1$ that bid $b'_1 = \frac{5}{4}, b'_2 = \frac{3}{4}$. This is a 2-SC that brings the joint utility of the two bidders and the miner from $0 \frac{1}{4}$. Under the Salsa Decomposition, the two steps are $(1, 1) \rightarrow (5/4, 1) \rightarrow (5/4, 3/4)$, and the second step is a 2-SC (the joint utility changes from $\frac{1}{8}$ to $\frac{1}{4}$ (recall that for each decomposed step, we consider the beginning of the step to be the true values and the end of the step to be the collusion bids). The other possible decomposition is $(1, 1) \rightarrow (1, 3/4) \rightarrow (5/4, 3/4)$, where no step is a 2-SC.

Lemma 3.10 (The “Salsa Dance” Lemma). *Every collusion from setting A to B has a one-by-one decomposition given by Definition 3.7, so that the following holds:*

- The initial setting is A and the final setting is B , and in the decomposition each bidder only changes their bid once.

- At steps where bidders in U_O change their bids, the bidder that changed their bid is unconfirmed after the step.
- At steps where bidders in D_I change their bids, the bidder that changed their bid is confirmed in the subsequent step.

PROOF. The first proposition holds by our explicit construction.

For the second proposition, examine the implications of the monotonicity Lemma 3.5. If a bidder in U_O is confirmed at some point during the stage where bidders in U_O raise their bids, then the property that some bidder in U_O is confirmed is maintained until the end of the process. However, by the definition of U_O , no bidder in U_O is confirmed at B , which is the last element of SEQ , so this is a contradiction.

This further implies that all bidders in U_O are unconfirmed throughout the stage where bidders in U_O raise their bids. By the non-bossy Lemma 3.4 together with the consistent tie-breaking assumption, this means that the number of confirmed bidders remains the same throughout that stage, and so at the end of the decreasing stage by bidders in D_I it is the same amount as in B .

For the third proposition, assume towards contradiction that a bidder in D_I decreases their bid and is then unconfirmed. Similarly (in an opposing way) to the U_O case, this implies that by the end of the decreasing stage there is a bidder in D_I which is unconfirmed. However, we previously concluded that during the last stage where bidders in U_O raise their bids, the number of confirmed bidders remains the same, in contradiction to all bidders in D_I being non-confirmed in B . $\square$

Lemma 3.11 (Single Mover Reduction). *Given a one-by-one decomposition with the properties of Lemma 3.10, there are two subsequent steps in the sequence so that there is a beneficial collusion from the preceding step to the subsequent step for the coalition C , where a single bidder changes their bid.*

PROOF. We assume towards contradiction that there are no two subsequent steps that allow a beneficial collusion between them. We can write the condition for the miner and the coalition C that prohibits the collusion for every two subsequent steps. We also write the condition of the original collusion (that gains a benefit of δ). We show that summing all the inequalities, and the collusion equality with the term δ , leads to a contradiction.

We introduce the following previously introduced notations: Let A and B be the settings of the collusion Ω_o , and let $X^1, \dots, X^k$ be all the intermediate steps of the decomposition SEQ .

When we argue for some bidder i , let X_{before}^i be the setting before it changes its bid, and X_{after}^i after. Since bidder i is the only bidder that changes their bid in that step, it holds that $u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) = u_i(X_{after}^i; X_{after}^i) - u_i(X_{after}^i; X_{before}^i)$. We note the following:

- For every bidder $i \in D_I$, by Lemma 3.10, the bidder is confirmed in X_{after}^i . Thus,

$$u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) = b'_i - b_i.$$

- For every bidder $i \in U_O$, by Lemma 3.10, the bidder is unconfirmed in X_{after}^i , and thus

$$u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) = 0.$$

- For every bidder $i \in U_I$, it is either confirmed or not confirmed in X^2 , and we have

$$\begin{aligned} & u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) \\ & \leq \max\{b'_i - p_i^{X^2} - (b_i - p_i^{X^2}), 0\} = b'_i - b_i. \end{aligned}$$

- For every bidder $i \in D_O$, it is either confirmed or not confirmed in X^2 , and we have

$$\begin{aligned} & u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) \\ & \leq \max\{b'_i - p_i^{X^2} - (b_i - p_i^{X^2}), 0\} = 0. \end{aligned}$$

Now, we can write the SCP inequalities:

$$\begin{aligned} & u_{\min}(A) + u_C(A; A) \geq u_{\min}(X^1) + u_C(X^1; A), \\ & u_{\min}(X^1) + u_C(X^1; X^1) \geq u_{\min}(X^2) + u_C(X^2; X^1), \\ & \dots \\ & u_{\min}(X^k) + u_C(X^k; X^k) \geq u_{\min}(B) + u_C(B; X^k), \\ & u_{\min}(B) + u_C(B; A) = u_{\min}(A) + u_C(A; A) + \delta. \end{aligned} \tag{8}$$

First, notice that all $u_{\min}$ terms cancel out between the LHS and RHS when summing all the inequalities, and $u_C(A; A)$ on both sides cancel as well. Let i^* be the last agent in $U \cup D$ that changes their bid in the sequence (i.e., $X_{after}^{i^*} = X^k$). We thus get, when summing all the inequalities:

$$\begin{aligned} & u_C(B; A) - u_C(B; X_{after}^{i^*}) \\ & + \sum_{i \in U \cup D \setminus \{i^*\}} u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) \geq \delta, \end{aligned} \tag{9}$$

Then, using our conclusions for each bidder type, we get:

$$\begin{aligned} & u_C(B; A) - u_C(B; X_{after}^{i^*}) + \sum_{i \in (U_I \cup D_I) \setminus \{i^*\}} (b'_i - b_i) + \sum_{i \in D_O \cup U_O} 0 \\ & \geq u_C(B; A) - u_C(B; X_{after}^{i^*}) \\ & + \sum_{i \in U \cup D \setminus \{i^*\}} u_C(X_{after}^i; X_{after}^i) - u_C(X_{after}^i; X_{before}^i) \geq \delta. \end{aligned} \tag{10}$$

Let $\sigma = U_I \cup D_I$. Since exactly the bidders in σ are confirmed in B , we have:

$$\begin{aligned} & u_C(B; A) = \sum_{i \in \sigma} (b_i - p_i^B), \\ & u_C(B; X_{after}^{i^*}) = \sum_{i \in \sigma \setminus \{i^*\}} (b'_i - p_i^B) + 1[i^* \in \sigma] \cdot (b_{i^*} - p_{i^*}^B) \end{aligned} \tag{11}$$

Thus, we can continue Eq. 10 with:

$$\begin{aligned} & 0 \stackrel{(*)}{=} \sum_{i \in \sigma} (b_i - p_i^B) - \left(\sum_{i \in \sigma \setminus \{i^*\}} (b'_i - p_i^B) + 1[i^* \in \sigma] \cdot (b_{i^*} - p_{i^*}^B) \right) \\ & + \sum_{i \in \sigma \setminus \{i^*\}} (b'_i - b_i) \\ & = u_C(B; A) - u_C(B; X_{after}^{i^*}) + \sum_{i \in \sigma \setminus \{i^*\}} (b'_i - b_i) + \sum_{i \in \sigma} 0 \geq \delta. \end{aligned} \tag{12}$$

The proof of (*) is a matter of case-analysis and we show it in the full version. We conclude that Eq. 12 shows $0 \geq \delta$, but since $\delta > 0$, this is a contradiction, and there must be some two subsequent steps constituting a beneficial collusion. $\square$

Lemma 3.12 (Single-Jump Property of Single Mover Collusion Coalition Value). *In a 2-SCP mechanism with a collusion where a single bidder i increases their bid from b_i to b'_i and a coalition C gains Δ in value, there is a single value b_i^* , so that for any small enough $\epsilon > 0$, there is a collusion for coalition C where only bidder i increases their value from $b_i^* - \frac{\epsilon}{2}$ to $b_i^* + \frac{\epsilon}{2}$, and the coalition gains Δ in value.*

A similar statement holds when the single mover decreases their value:

Lemma 3.13. *In a 2-SCP mechanism with a collusion where a single bidder i decreases their bid from b_i to b'_i and a coalition C gains Δ in value, there is a single value b_i^* , so that for any small enough $\epsilon > 0$, there is a collusion for coalition C where only bidder i decreases their value from $b_i^* + \frac{\epsilon}{2}$ to $b_i^* - \frac{\epsilon}{2}$, and the coalition gains Δ in value.*

Finally, we can reduce any SCP to one with a single mover, and show by the single-jump property that we can gain a fixed value of Δ for the coalition while losing at most ϵ for the moving bidder, and can further isolate a single bidder, possibly different than the mover, that gains and allows a 2-SCP.

Theorem 3.14 (One Mover, One Beneficiary). *The class of 2-SCP mechanisms equals the class of SCP mechanisms.*

4 Corollaries Of The Main Characterization

We present two applications of the above characterization.

Lemma 4.1. *With a block size of 1 (equivalently, a single-item auction), the class of 2-SCP TFM's is exactly the class of first-price auctions with reserve r that is burned.*

PROOF. (2-SCP $\implies$ first-price with burned reserve)

Consider any 2-SCP auction a, p, β . By our characterization, it is also SCP. [15] show that SCP implies Global-SCP, and provide a characterization of Global-SCP auctions in the single-item case (notice that this means SCP implies the characterization, and we do not need to worry about Global-SCP, which we did not define). The characterization is as follows: For some reserve price r , the auction confirms the item to the highest bidder above r , and burns r . The payment function may be determined arbitrarily (as long as it respects individual rationality and burn-balance).

Now, consider the payment function for our auction, which is not only Global-SCP, but SCP. The function can not depend on losing bids: If it does, there is an SCP of the losing bids to increase the payment, which benefits the miner (and does not harm the losing bids). Thus, there is $f : R_+ \rightarrow R_+$ so that $p(\mathbf{b}) = f(\max \mathbf{b})$, i.e., the payment only depends on the winning bid. Now, assume towards contradiction that for some v , $f(v) < v$. Consider some $f(v) < v' < v$, and let $\mathbf{b} = (v, v')$ (call it setting A). Consider the 1-SCP between the miner and bidder 2 where bidder 2 bids $2 \cdot v$ (call it setting B). We have $u_{\text{miner}}(A) + u_2(A; A) = f(v) - r$, where r is the reserve price that is burned. We also have $u_{\text{miner}}(B) + u_2(B; A) = v' - r > f(v) - r$, and so this is a 1-SC. We conclude that we must have $f(v) = v$.

(first-price with burned reserve $\implies$ SCP)

If all bidders bid honestly, we have

$$u_{\text{miner}}(\mathbf{b}) = 1[\max \mathbf{b} \geq r] \cdot (\max \mathbf{b} - r),$$

and $u_i(\mathbf{b}) = 0$ for any bidder i . Under any collusion, if bidder $i \in C$ ends up the winner, the joint utility of the collusion is $b_i - r$. If no bidder in the coalition wins, the coalition utility is either $p_i - r$ for some $i \notin C$, or 0. In any case, this is less than $1[\max \mathbf{b} \geq r] \cdot (\max \mathbf{b} - r)$, and so the SCP condition holds. $\square$

This characterization is interesting as it shows that the current popular implementations (Bitcoin's pay-as-bid auction, and Ethereum's EIP-1559, which is essentially a first-price auction with burned reserve) fit exactly the class of collusion-free auctions, in the SCP sense. Moreover, in the single-item case, the 2-SCP characterization can be tightened to 1-SCP.

Lemma 4.2. *With a block size 1, 1-SCP $\implies$ 2-SCP.*

PROOF. We prove the contrapositive, i.e., we show that if 2-SCP is violated, then so is 1-SCP. Consider a beneficial 2-SCP collusion Ω , going from setting A to B . By Lemma 3.11, we can assume there is a single bidder i who changes their bid. If the other bidder in the collusion j does not win the item under B , then their utility in B is 0, and by individual rationality of A , they do not gain from the collusion. Thus, the 1-SC that only involves the miner and agent i is beneficial as well.

Otherwise, the winner in B is bidder j . Then, the collusion can be implemented without the involvement of bidder i : Instead of bidder i changing their bid (from b_i^A to b_i^B), the miner drops their bid b_i^A , and introduces a fake bid b_i^B . Call this collusion, that involves only bidder j , Ω' . Now, notice that the utility of the miner and bidder j under B with is the same as the utility of the miner and bidders i, j under B (since i has utility 0). The utility of the miner and bidder j under A is, by individual rationality, at most the utility of the miner and the bidder i, j under A . Thus, if Ω is beneficial for the colluders, then so is Ω' . $\square$

The corollary immediately follows from the two lemmas:

Corollary 4.3. *With a block size of 1 (equivalently, a single-item auction), the class of 1-SCP TFM's is exactly the class of first-price auctions with reserve r that is burned.*

In [7], they show an impossibility for UIC + 1-SCP mechanisms with a finite block size. However, with an infinite block size, [6] present the discount auction, which is UIC+1-SCP. We complement these results by characterizing UIC+2-SCP mechanisms in the infinite block-size case:

DEFINITION 4.4. *Fully-burned Posted-price Mechanisms are the class of mechanisms where for some constant r , all bidders with $b_i \geq r$ are confirmed, they each pay r , and all payments are burned.*

Theorem 4.5. *The class of UIC + 2-SCP mechanism for infinite block size is exactly the class of fully burned posted-price mechanisms.*

5 Discussion

Our work shows that the class of 2-SCP mechanisms is the same as the class of SCP mechanisms. This, in the view of the miner, means that when a mechanism is subject to collusion, they only need to find 2 co-conspirators to implement a beneficial one. This shows that in some sense of *coalitional* complexity, collusion is simple.

One non-trivial assumption we make is that of consistent tie-breaking towards *zero-utility* bidders w.r.t. unconfirmed bidders. Let us describe an auction which violates this assumption, and for which our single-mover argument fails.

Example 5.1. Consider an auction where payments are fully burned. A winning bid is only confirmed if there is a losing bid with value at least 8. If a bid with value at least 10 exists, all bids of value 8 and more are confirmed and pay 6.5.

Consider two bidders $b_1 = 10, b_2 = 1$ (call this setting A), that collude to change their bids to $b'_1 = 9, b'_2 = 8$ (call this setting B). This is a beneficial collusion of the miner and two bidders as their joint utility increases from 0 to 1. However, no single-move decomposition of the collusion is beneficial. Let X be the setting with bids 9, 1, and Y the setting with bids 10, 8. The joint utility of the coalition moving from A to Y goes from 0 to -2 . From A to X it goes from 0 to 0. From X to B it goes from 0 to 0, and from Y to B it goes from 5 to 1.

Going beyond our discussion of collusion size, there are other aspects of the collusion that may still be hard. In terms of *computational* complexity, it may be hard to identify how to implement a successful collusion. This depends on the computational model of how the problem is represented. Importantly, at least at surface-level, it seems like our result does not change the basic complexity. Consider two natural choices. If we assume that the problem is given as a full discrete table mapping all possible types (values) to outcomes (confirmation, payment, and burn), then the problem of finding a collusion is a matter of iterating over all possible types, and finding two settings where a beneficial collusion exists. This is polynomial (quadratic) in the input size, and true even without our reduction. On the other hand, if one assumes as in [19] that the problem is given as a Boolean circuit, then the problem of finding a beneficial 2-SCP is coNP-complete.

DEFINITION 5.2. *Boolean circuit representation of a, p, β :* Consider a finite set of values $v^1, \dots, v^k$, and let $\bar{v} \in \{0, 1\}^{\log k}$ be binary representation of them. With n bidders, let $C_1^a, \dots, C_n^a$ be Boolean circuits that receive n boolean vectors of size $\log k$ (so overall $n \log k$ bits), where each C_i^a outputs $a_i(v_1, \dots, v_n) \in \{0, 1\}$. $C_1^p, \dots, C_n^p, C_1^\beta, \dots, C_n^\beta$ each output a binary representation of the payment and burn for each bidder (which we assume, for simplicity, are in $v^1, \dots, v^k$ as well).

We are interested in the search problem of finding a 2-SC (or reporting there is none, if none exists). To understand its complexity, we can define the decision problem of whether the auction is 2-SCP:

DEFINITION 5.3. *2-SCDP (2-SCP Decision Problem):* Given circuits $C_1^a, \dots, C_n^a, C_1^p, \dots, C_n^p, C_1^\beta, \dots, C_n^\beta$ that represent an individually rational, burn-balanced single-item auction, decide whether the auction represented by the circuits is 2-SCP.

The problem is in coNP, since a witness to show that it is *not* 2-SCP is simply two sets of bidder values $v_1, \dots, v_n, v'_1, \dots, v'_n$, and

(at most) two bidder indices i, j . Given this polynomial-size witness, a beneficial 2-SC can be verified in polynomial time.

Lemma 5.4. *2-SCDP is coNP-complete.*

In terms of *communication* complexity, it is important to understand how many rounds of communication are needed between the colluders to establish the collusion, given that the values of the bidders are private information. This is especially important when the collusion coordination must be very quick, e.g., Ethereum mines a block every 12 seconds on average [35]. Ideally (for the colluders), no communication would be needed, and a *tacit* collusion strategy exists that each may apply on their own. It is clear that at most two additional rounds of communication are required for a collusion: One round where all bidders share their values with the miner, another one where the miner tells each bidder what value to bid, and then the mandatory round where all bidders make their bids. It is interesting to understand in what settings the first two rounds can be avoided. Moreover, this protocol leaves all the bargaining power in the hands of the miner, and this may not be acceptable for the bidders. It is interesting to know the “price of bargaining” in terms of communication, if one was to instead follow an established process of bargaining [8, 30].

5.1 Open questions.

We list some remaining open questions.

- *Removing the consistent-tie-breaking assumption.* Although we find the assumption natural, we would ideally wish to avoid limiting the range of auction formats we consider. Moreover, it remains possible that a non consistent-tie-breaking auction *does* separate 2-SCP from higher c -SCP, and this could be interesting for collusion prevention.
- *A separation of 1-SCP and 2-SCP with a finite block size greater than 1.* The discount auction [6] shows a separation in the infinite block size case. Our Lemma 4.2 shows they are the same with a block size of 1. What happens in-between?
- *Characterizing 2-SCP with finite block size greater than 1.* Our Lemma 4.1 shows a characterization of 2-SCP mechanisms with $k = 1$. For general k , a good starting point is the Global-SCP characterization for general k [13]. An argument similar to Lemma 4.1, where the miner could work together with losing bids, would show that winning bids pay at least the value of the lowest winning bid.
- *A Global-SCP mechanism which is 1-SCP, but not 2-SCP.* This is an open question of [15] that remains open. Global-SCP is weaker than SCP, so this is not ruled out by our results.
- *Characterizing SCP and Global-SCP, and applying our techniques beyond auctions.* The TFM literature is focused on auctions, but the collusion notions are general and can apply to other mechanism design settings. Moreover, we show it is possible to simplify a collusion notion by decomposition, isolating a single agent that makes a change, and one other agent that benefits from it. This may (or may not) hold more generally in settings such as bribing and censorship attacks [4, 10], or post-MEV environments [2].

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